

IF Function in Excel

1. What is IF?

The IF function tests a condition and returns one result if the condition is TRUE and a different result if it is FALSE. It solves the problem of making a decision for every row of data, such as labelling a student as Pass or Fail. A Data Analyst uses it to categorize records, flag important rows, and create new columns from existing data.

2. Syntax

`=IF(logical_test, value_if_true, value_if_false)`

Argument	Required?	What it accepts
logical_test	Required	A condition that is either TRUE or FALSE, such as <code>B2>=50</code> or <code>C2="Sales"</code>
value_if_true	Optional*	What to return when the test is TRUE: text in quotes ("Pass"), a number, a cell, or a calculation
value_if_false	Optional*	What to return when the test is FALSE: the same types as above

*Excel allows these two to be left out, but beginners should always fill in both. Comparison operators used in the test: `=` equal, `<>` not equal, `>` greater, `<` less, `>=` greater or equal, `<=` less or equal.

3. Simple Example

Check whether each student passed (pass mark is 50):

	A	B	C
1	Student	Marks	Result
2	Rahim	75	Pass
3	Karim	45	Fail
4	Hasan	82	Pass
5	Nabil	55	Pass

Formula in C2: `=IF(B2>=50, "Pass", "Fail")`, then copied down to C5.

For Rahim, Excel checks whether 75 is 50 or more. This is TRUE, so it returns "Pass". For Karim, 45 is not 50 or more. This is FALSE, so it returns "Fail". Excel repeats this for each row.

4. Another Example

Pay a 5% bonus on sales of 10000 or more, otherwise 0:

	A	B	C
1	Employee	Sales	Bonus
2	Anika	12000	600
3	Sabbir	8500	0
4	Mitu	15000	750
5	Tanvir	9800	0

Formula in C2: `=IF(B2>=10000, B2*5%, 0)`, then copied down to C5.

For Anika, 12000 is 10000 or more, so Excel returns the calculation $12000 \times 5\% = 600$. For Sabbir, 8500 is below 10000, so Excel returns 0.

5. When to Use It

- Label exam marks as Pass or Fail
- Check whether a sales target was met or not met
- Decide bonus eligibility based on sales or performance
- Mark attendance as Eligible or Not Eligible
- Flag high-value orders or overdue payments for review

6. Important Things to Remember

- IF checks one condition and returns one of two results: the TRUE result or the FALSE result.
- Text results must be inside quotation marks ("Pass"). Numbers, cell references, and calculations do not need quotes.
- The results can be calculations, not just labels, for example `B2*5%`.
- If you leave out `value_if_false`, as in `=IF(B2>=50, "Pass")`, Excel shows FALSE for failing rows. Always provide both results.
- Text comparisons are not case-sensitive: `C2="sales"` matches "Sales".
- IF works on one row at a time. Write it in the first row, then copy it down the column. For more than two outcomes, one IF can be placed inside another.

7. Common Mistakes

Mistake: Forgetting quotation marks around a text result, which can cause a `#NAME?` error. **Fix:** Write "Pass" and "Fail" with quotes.

Mistake: Using the wrong operator at the boundary, such as `>50` instead of `>=50`, so a student with exactly 50 fails. **Fix:** Decide whether the boundary value should count, then choose `>` or `>=`.

Mistake: Putting a number in quotes inside the test, such as `B2>="50"`. Excel treats it as text and the result is wrong. **Fix:** Write numbers without quotes: `B2>=50`.

Mistake: Missing or extra brackets and commas. **Fix:** Check the formula bar; Excel colors each pair of brackets to help you see which ones match.

8. Data Analyst Tip

Put the condition value in its own cell instead of typing it inside the formula. For example, store the pass mark 50 in E1 and use `=IF(B2>=$E$1, "Pass", "Fail")`. If the pass mark changes, update E1 once and every result updates. IF is also useful for creating a status column that you can then filter, sort, or count.